Autoresearch: Engineering an Auditable Agentic Research Loop for Event-Driven Earnings Prediction

A Case Study in Quantitative System Design, Out-of-Sample Empirical Validation, and Anti-Overfitting Governance

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ABSTRACT

Quantitative strategy discovery faces a fundamental tension between research velocity and statistical validity: manual hypothesis testing is slow and inconsistent, whereas unconstrained automated search rapidly overfits historical noise. This paper presents **Autoresearch**, an agent-assisted quantitative research architecture that couples bounded hyperparameter and feature search with automated critique, strict walk-forward validation, and mandatory human promotion gates. We examine the application of this framework to event-driven corporate earnings prediction—specifically, forecasting the direction and magnitude of post-announcement equity price movements across industry cohorts. Evaluating an out-of-sample validation dataset of N = 1,197 corporate earnings events across eight industry cohorts, we find an aggregate directional accuracy of 50.7% (95% Wilson confidence interval: 47.9%–53.5%), a receiver operating characteristic area under the curve (ROC AUC) of 0.503, a mean absolute magnitude error (MAE) of 8.60 percentage points, and an out-of-sample move R-squared of -5.614. Rather than establishing a deployable trading edge, the empirical audit exposes three actionable failure mechanisms: (i) severe magnitude overprediction in the top confidence quintile (20.1% predicted vs. 6.0% realized move) and top signal-strength quintile (24.2 percentage points MAE; Spearman rho = -0.367), (ii) performance dilution from a generic global fallback model (48.4% accuracy across 50.7% of events) relative to specialized industry models such as Software (54.5% accuracy, n = 391), and (iii) statistically significant underperformance in Biotechnology (41.1% accuracy, n = 124, unadjusted p = 0.038). We demonstrate how an auditable validation harness transforms weak aggregate forecasts into targeted engineering interventions—prioritizing post-hoc probability calibration, robust quantile loss functions, immutable point-in-time prediction ledgers, and multiple-testing controls before scaling agent autonomy or committing capital.

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1. Introduction: Automating Quantitative Discovery Without Automating False Confidence

Systematic quantitative research follows an iterative scientific cycle: formulate economic or technical hypotheses, engineer candidate predictors, estimate model parameters, evaluate out-of-sample behavior, diagnose failure modes, and refine the specification. In practice, manual execution of this loop suffers from cognitive bottlenecks and inconsistent record-keeping. Conversely, naive automation—allowing machine learning pipelines or large language model (LLM) agents to search unconstrained feature and parameter spaces—inevitably manufactures backtest overfitting and false discoveries that collapse out of sample [1, 2, 3]. As Bailey et al. (2014) and Harvey, Liu, and Zhu (2016) demonstrate, repeated trials across historical financial series guarantee the appearance of high-Sharpe or high-accuracy configurations purely by chance unless search complexity and selection bias are explicitly penalized [1, 2, 3].

Autoresearch addresses this tension by structuring quantitative discovery as a bounded, auditable state machine inspired by recent advances in reasoning-and-acting (ReAct) language agents [4], verbal self-reflection loops (Reflexion) [5], and automated scientific discovery architectures [6, 7]. Rather than treating an LLM as an unconstrained code generator, the framework constrains agent autonomy within explicit trial budgets, feature-complexity caps, rolling walk-forward evaluation windows, and mandatory human researcher checkpoints. While the core state-machine architecture applies broadly across systematic futures and multi-asset technical strategies, its most demanding testbed is event-driven corporate earnings prediction.

Corporate earnings announcements represent discrete, high-variance information shocks that drive both immediate price jumps and Post-Earnings-Announcement Drift (PEAD) [8, 9]. However, building a profitable event-driven strategy requires solving three distinct problems that are frequently conflated in empirical machine learning: (1) Fundamental Surprise Prediction (forecasting whether reported revenue and EPS will exceed consensus expectations), (2) Return Distribution Forecasting (estimating the direction and absolute magnitude of the post-announcement equity price reaction given pre-announcement expectations already embedded in market prices), and (3) Positive-Expectancy Trade Execution (translating a statistical forecast into positive net risk-adjusted returns after accounting for bid-ask spreads, market impact, options implied-volatility buildup, and the sharp post-announcement implied-volatility crush documented by Patell and Wolfson, 1979, and Dubinsky et al., 2019) [10, 11].

This paper documents both the architectural design of the Autoresearch system and the empirical out-of-sample validation of its pre-earnings forecasting pipeline across 1,197 corporate earnings events. By strictly separating aspirational system architecture from verified empirical performance, we illustrate why independent validation harnesses are the foundational prerequisite for agentic quantitative finance.

2. System Objectives and Governance Criteria

An institutional automated research platform must satisfy three interdependent classes of objectives. Predictively, it seeks statistically significant directional ranking above chance (ROC AUC > 0.50), well-calibrated directional probability estimates, and accurate conditional expectation or quantile estimates of absolute post-earnings price moves across heterogeneous industry groups [12, 13]. Operationally, it aims to automate repetitive feature screening, hyperparameter tuning, and diagnostic summarization via specialized software agents, while persisting every trial—successful or failed—into an immutable analytical store [6, 7]. Governance requirements include enforcing strict structural complexity limits (e.g., capping active predictors per strategy), eliminating look-ahead and survivorship bias through point-in-time data alignment, penalizing in-sample to out-of-sample degradation, and requiring explicit human sign-off prior to live capital allocation [2, 14].

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A recurring governance challenge in multi-strategy research is harmonizing target metrics across distinct asset classes and holding periods. Within the Autoresearch framework, different modules specify distinct quantitative promotion hurdles. For high-frequency and intraday futures strategies, the design specifies a walk-forward win rate of at least 75%, an annualized Sharpe ratio of at least 2.0, maximum drawdown of at most 10%, minimum trade frequency of 30 trades per year, and an out-of-sample to in-sample performance retention ratio of at least 0.80 [2, 15]. For the event-driven equity earnings module, the roadmap specifies a backtest directional win rate of at least 60% and a payoff ratio of at least 1.30 at the strategy level, alongside statistical forecast gates requiring a directional ROC AUC of at least 0.70 and an expected-move regression R-squared of at least 0.40.

In our empirical audit below, we treat these figures strictly as prospective acceptance hurdles rather than achieved performance, and we demonstrate why forecast-level metrics (ROC AUC, MAE, R-squared, Brier score) must be evaluated in a dedicated validation gate before any strategy-level Sharpe or win-rate claim is credited.

3. Architecture: Bounded Agentic Discovery, Cohort Specialization, and Independent Validation

As illustrated in Figure 1, the Autoresearch framework is organized into four decoupled stages that separate hypothesis generation from empirical verification. At the core of the discovery loop is a directed state-machine orchestrator (LangGraph / ReAct topology) [4, 7] that coordinates four specialized roles: (i) an Optimizer Agent that conducts Bayesian hyperparameter optimization using Optuna's Tree-structured Parzen Estimator (TPE) and Covariance Matrix Adaptation Evolution Strategy (CMA-ES) samplers [16, 17, 18] under strict trial budgets (at most 200 trials on local two-year windows and 300 trials on global five-year windows); (ii) a Backtester Agent that executes rolling walk-forward simulations [14, 15]; (iii) an Indicator and Feature Discovery Agent that evaluates candidate predictors using permutation importance [19] subject to a hard cap of five active indicators; and (iv) a Structured LLM Critic Agent that diagnoses regime-dependent weaknesses and proposes a single testable next experiment [4, 5].

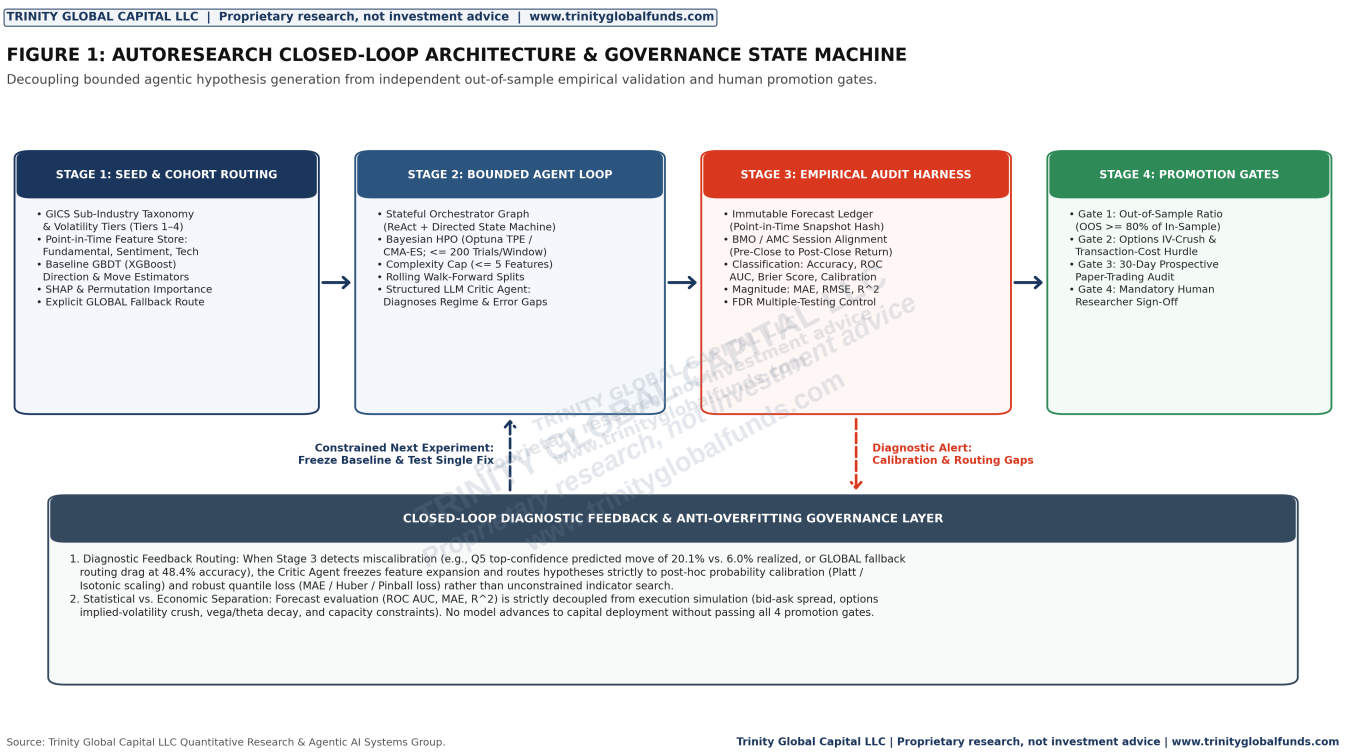


Figure 1: Autoresearch Closed-Loop Architecture and Anti-Overfitting Governance State Machine.

Corporate earnings reactions exhibit pronounced cross-sectional heterogeneity: a 10% revenue surprise in high-multiple enterprise software carries fundamentally different valuation elasticity and implied-volatility dynamics than an equivalent surprise in regulated utilities or clinical-stage biotechnology. Relying on a single pooled cross-sectional model forces homogeneous parameter weights across incompatible economic regimes. To address this, the earnings prediction architecture partitions the equity universe using the Global Industry Classification Standard (GICS) sub-industry taxonomy—shown by Bhojraj, Lee, and Oler (2003) to dominate alternative classification schemes in explaining cross-sectional return comovement and valuation multiples [20]—combined with four historical earnings-volatility tiers. For each cohort, baseline Gradient Boosted Decision Trees (XGBoost) [21] estimate both directional probability and expected absolute move magnitude, accompanied by SHAP (SHapley Additive exPlanations) values [22] to attribute predictions across fundamental surprise history, revisions, and technical setup features. Events lacking a dedicated sub-industry model are routed to a pooled GLOBAL fallback model.

Crucially, the empirical evaluation pipeline is engineered as a standalone, model-agnostic audit harness. It ingests archived point-in-time pre-earnings forecasts, retrieves realized market prices via an embedded analytical database

(DuckDB) [23] and adjusted market data feeds, aligns announcement timing, computes parametric and non-parametric diagnostics, and exports reproducible audit artifacts. This decoupling guarantees that forecast quality is graded independently of the agent loop that produced it.

4. Empirical Methodology: Event Alignment and Statistical Metrics

Each observation in the validation study corresponds to a unique firm-announcement pair (s, t), where s denotes the equity ticker and t denotes the earnings announcement date. For every event, the validation pipeline extracts the pre-announcement record containing the routed model source (either a specialized GICS cohort model or the GLOBAL fallback model), the predicted direction (assigned as UP when the estimated upward probability is at least 0.50, and DOWN otherwise), directional confidence (distance from the uninformative 0.50 prior), predicted move magnitude, and composite signal strength (0 to 100).

To prevent look-ahead bias and ensure exact alignment between the earnings release and the market session in which prices first react, realized returns are aligned according to announcement timing: for After-Market-Close (AMC) announcements, the pre-event baseline price is the closing price on announcement date t and the post-event price is the closing price on the next trading session t+1; for Before-Market-Open (BMO) announcements, the pre-event baseline price is the closing price on session t-1 and the post-event price is the closing price on announcement date t. The realized signed one-session return is r = (P_post / P_pre) - 1, and the realized move magnitude is m = |r|. Realized direction is mapped as UP if r > 0 and DOWN if r <= 0.

We evaluate N = 1,197 deduplicated corporate earnings events across an out-of-sample evaluation window. Table 1 summarizes the headline directional and magnitude performance across the full sample, and Table 2 reports the directional confusion matrix and class-conditional precision and recall.

Table 1: Out-of-Sample Aggregate Forecast Evaluation (N = 1,197 Corporate Earnings Events)

Evaluation Metric	Observed Value	Benchmark / Target	Diagnostic Interpretation
Evaluated Earnings Events (N)	1,197	—	Unique deduplicated (Symbol, Announcement Date) pairs
Directional Accuracy	**50.71%**	**> 50.0% (60.0% Target)**	**607 correct calls; inside 95% Wilson CI (47.9%–53.5%)**
Naive Majority Baseline (Always-UP)	51.29%	614 / 1,197 actual UP	Model underperforms a trivial always-long baseline by -58 bps
Directional ROC AUC	**0.503**	**>= 0.700 Target**	**Indistinguishable from random coin-flip ranking (0.500)**
Top-Quintile Confidence Accuracy (Q5)	48.75%	> 50.71% Overall	High model confidence fails to improve directional hit rate
Signed Return Correlation	-0.004	> 0.000	Zero linear explanatory power on signed event returns
Magnitude Mean Absolute Error (MAE)	**8.60 pp**	**Lower is better**	**Median absolute error is 3.07 pp (right-skewed tail errors)**
Magnitude Root Mean Squared Error	17.44 pp	Lower is better	Severe sensitivity to extreme magnitude overpredictions
Magnitude Out-of-Sample R-Squared	**-5.614**	**>= 0.400 Target**	**Performs worse than predicting the sample mean move**
Within +/-50% / +/-100% of Actual	27.2% / 64.2%	—	Only ~1/4 of magnitude forecasts fall within +/-50% band

Table 2: Out-of-Sample Directional Confusion Matrix and Class Precision/Recall (N = 1,197)

Actual \ Predicted	Predicted UP (p >= 0.5)	Predicted DOWN (p < 0.5)	Row Total	Class Recall
Actual UP (r > 0)	289	325	614 (51.3%)	47.07%
Actual DOWN (r <= 0)	265	318	583 (48.7%)	54.55%
Column Total	554 (46.3%)	643 (53.7%)	1,197 (100.0%)	—
Class Precision	**52.17%**	**49.46%**	**—**	**Accuracy: 50.71%**

Using the Wilson (1927) score interval for a binomial proportion [24], the approximate 95% confidence interval for the model's 50.71% directional accuracy under event independence is [47.88%, 53.54%]. Because earnings announcements within the same calendar week share common macro and sector shocks, cross-sectional correlation widens this confidence band further. Moreover, a trivial majority-class classifier that predicts UP for every announcement achieves 614 / 1,197 = 51.29%, outperforming the model by 58 basis points. Consequently, the aggregate model exhibits no statistically defensible directional edge.

5. Diagnostic Decomposition of Failure Modes and Evidentiary Limitations

The primary value of a structured validation harness is not merely assigning a pass/fail grade, but isolating why an aggregate model fails. Disaggregating the 1,197 events across confidence quintiles, signal-strength quintiles, and GICS industry cohorts (Figure 2 and Tables 3–4) reveals three distinct structural failure modes.

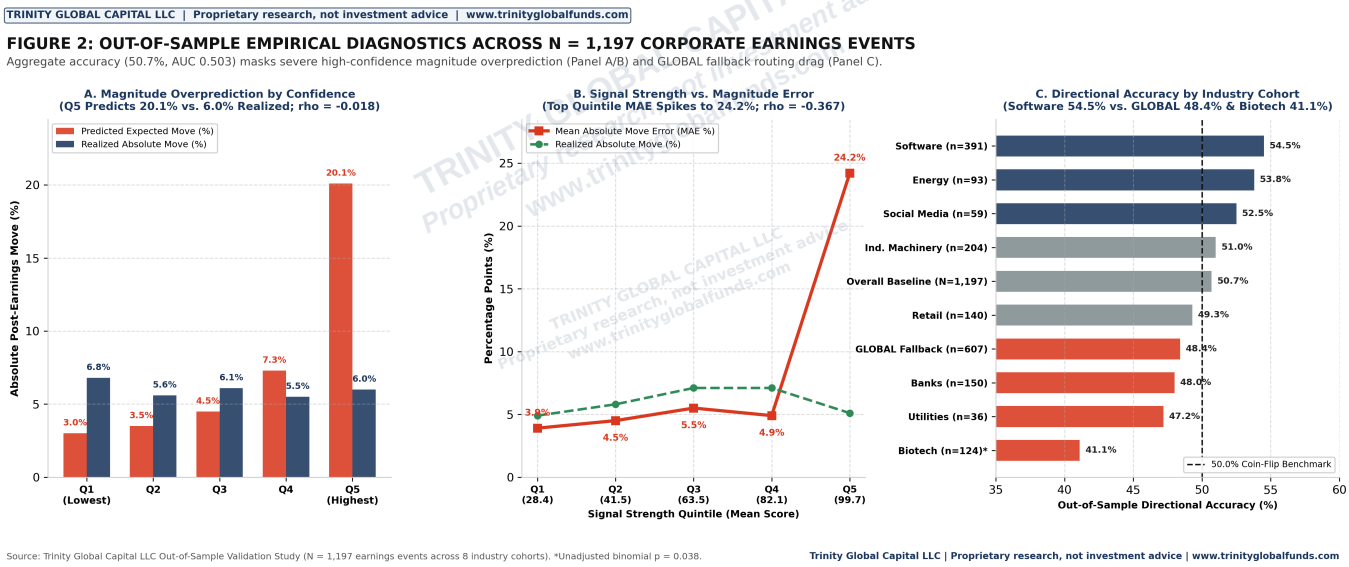


Figure 2: Out-of-Sample Empirical Diagnostics Across N = 1,197 Corporate Earnings Events.

Failure Mode 1 — Severe Magnitude Miscalibration in High-Conviction Tails: Whereas realized post-earnings absolute moves remain stable across all five confidence quintiles—averaging between 5.5% and 6.8% (Table 3, Panel A)—the model's predicted move escalates non-linearly from 3.0% in Q1 (lowest confidence) to 20.1% in Q5 (highest confidence), against a realized Q5 average move of just 6.0%. A parallel distortion occurs across Signal Strength quintiles (Table 3, Panel B): as mean signal strength rises from 28.4 (Q1) to 99.7 (Q5), directional accuracy remains flat (49.2% vs. 53.3%, Spearman rho = 0.033), but magnitude MAE explodes from 3.9 percentage points in Q1 to 24.2 percentage points in Q5 (Spearman rank correlation between signal strength and negative absolute move error is rho = -0.367). Within the Retail cohort (n = 140), directional accuracy is near baseline (49.3%), but magnitude MAE reaches 35.3 percentage points (Table 4). Because standard squared-error (L2) boosting objectives heavily weight extreme

historical outliers, unclipped target transformations or unit-scaling discrepancies propagate directly into upper-tail predictions. Furthermore, while the Spearman correlation between directional confidence and realized absolute move size is rho = -0.018, classification probability calibration and return magnitude estimation are distinct statistical properties: proper verification of directional probabilities requires Brier (1950) skill scores [12] and reliability diagrams via Platt scaling or isotonic regression [13, 25], whereas magnitude estimation requires robust quantile regression (L1 / pinball loss) [26].

Table 3: Diagnostic Quintile Breakdown by Directional Confidence and Signal Strength (N = 1,197)

Quintile Bin	Events (n)	Mean Score	Dir. Accuracy	Realized Move	Predicted Move	Move Error (MAE)
Confidence Q1 (Lowest)	240	0.019	53.8%	6.8%	3.0%	5.1%
Confidence Q2	239	0.057	51.0%	5.6%	3.5%	4.8%
Confidence Q3	239	0.106	47.7%	6.1%	4.5%	5.4%
Confidence Q4	239	0.212	52.3%	5.5%	7.3%	6.9%
Confidence Q5 (Highest)	**240**	**0.535**	**48.8%**	**6.0%**	**20.1%**	**20.8%**
Signal Strength Q1 (Lowest)	240	28.4	49.2%	4.9%	3.2%	3.9%
Signal Strength Q2	239	41.5	48.1%	5.8%	4.1%	4.5%
Signal Strength Q3	239	63.5	53.1%	7.1%	6.0%	5.5%
Signal Strength Q4	239	82.1	49.8%	7.1%	6.8%	4.9%
Signal Strength Q5 (Highest)	**240**	**99.7**	**53.3%**	**5.1%**	**25.4%**	**24.2%**

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Failure Mode 2 — Cohort Heterogeneity, Multiple Testing, and Global Fallback Drag: Table 4 decomposes performance by GICS industry cohort and routed model source. Exactly 607 of the 1,197 events (50.7%) were routed to the generic GLOBAL fallback model, which achieved only 48.43% directional accuracy (AUC = 0.486). By contrast, the dedicated Software cohort model (n = 391) achieved 54.48% accuracy (AUC = 0.534) and Energy (n = 93) achieved 53.76% accuracy with a 2.8 percentage point MAE. Pooling heterogeneous industries into a single fallback estimator dilutes domain-specific signal. At the same time, while Biotechnology exhibits a nominal binomial p-value of 0.038 (41.13% accuracy across n = 124 events), testing eight cohorts simultaneously inflates the family-wise type-I error rate. Applying the Benjamini and Hochberg (1995) False Discovery Rate (FDR) procedure [27] across M = 8 cohorts yields an adjusted q-value of 0.304. Cohort anomalies must therefore be treated as diagnostic priors for out-of-sample testing—not proof that inverting Biotechnology signals or isolating Software signals guarantees live alpha [3].

Table 4: Out-of-Sample Performance by Industry Cohort and Routed Model Source (N = 1,197)

Cohort / Model Source	Events (n)	Share	Dir. Accuracy	vs. Overall	ROC AUC	Move MAE	p-val (Unadj.)	Diagnostic Assessment
Software (Dedicated)	**391**	**32.7%**	**54.48%**	**+3.77 pp**	**0.534**	**6.3%**	**0.143**	**Strongest cohort; positive lift**
Energy	93	7.8%	53.76%	+3.05 pp	0.516	2.8%	0.605	Modest lift; tightest MAE
Social Media (Dedicated)	59	4.9%	52.54%	+1.83 pp	0.546	5.5%	0.796	Small sample; exploratory
Industrial Machinery	204	17.0%	50.98%	+0.27 pp	0.501	5.3%	0.944	At random baseline
Retail (Dedicated)	**140**	**11.7%**	**49.29%**	**-1.42 pp**	**0.449**	**35.3%**	**0.736**	**Severe magnitude scaling bug**
Banks	150	12.5%	48.00%	-2.71 pp	0.508	3.0%	0.515	Low vol; below baseline
Utilities	36	3.0%	47.22%	-3.49 pp	0.489	2.8%	0.740	Small sample; below baseline
Biotechnology	**124**	**10.4%**	**41.13%**	**-9.58 pp**	**0.452**	**5.6%**	**0.038***	**Significant negative skill**
GLOBAL Fallback Route	**607**	**50.7%**	**48.43%**	**-2.28 pp**	**0.486**	**4.2%**	**—**	**Primary drag (>50% of events)**

Failure Mode 3 — Data Provenance and Event-Window Accounting: Our audit of the validation pipeline also identified three subtle data-engineering vulnerabilities common in production quant workflows: (i) selecting historical forecast batches by file modification timestamp rather than strict announcement-date intervals, (ii) deduplicating forecasts without cryptographically verifying that the prediction timestamp strictly precedes the announcement timestamp, and (iii) computing total evaluated event counts after performing an inner join between forecasts and realized market prices, which obscures pre-join coverage and missing-data attrition.

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6. Iterative Engineering: From Aggregate Failure to Targeted Remediation

Comparing sequential validation snapshots illustrates how an auditable loop guides research prioritization. An initial eight-week validation sweep (N = 1,150 events) recorded 51.0% directional accuracy, 0.514 ROC AUC, and 8.30 percentage points of magnitude MAE. Expanding the harness to N = 1,197 events with full quintile, cohort, and model-routing attribution confirmed that aggregate performance remained near baseline (50.7% accuracy, 0.503 AUC, 8.60 percentage points MAE), but isolated the exact mechanical drivers of error.

Without disaggregated diagnostics, a research team facing a 50.7% accuracy score typically defaults to adding more predictors—expanding the feature space and worsening overfitting [1, 2]. With diagnostic attribution, the engineering roadmap shifts entirely toward four structural remediations: (1) freeze the baseline predictor set and enforce cryptographic point-in-time timestamp verification alongside pre-join coverage accounting; (2) replace squared-error (L2) magnitude regression with robust quantile regression (L1 / pinball loss) [26] or Huber loss on volatility-standardized returns, eliminating the Retail (35.3% MAE) and Q5 (20.1% predicted vs. 6.0% realized) scaling blowups; (3) apply post-hoc probability calibration (isotonic regression or Platt sigmoid scaling) [13, 25] verified via Brier skill scores [12] on held-out folds before using confidence as a trade-sizing filter; and (4) decommission unspecialized GLOBAL fallback routing (48.4% accuracy) so that only validated industry cohorts are eligible for paper trading.

7. Blueprint for Scaling Scientific Discipline, Compute, and Capital

To transition Autoresearch from a diagnostic prototype into an institutional production platform, scaling must proceed across four sequential gates: Evidence Quality, Statistical Methodology, Computational Orchestration, and Economic Execution.

1. Evidence and Provenance Scaling: Replace file-based forecast archiving with an immutable prediction ledger backed by columnar analytical storage (Parquet / DuckDB) [23]. Every generated forecast must record a cryptographic SHA-256 hash of the training dataset, model binary version, exact UTC generation timestamp, point-in-time feature cutoff timestamp, and explicit GICS sub-industry integer code [20].

2. Statistical and Econometric Scaling: Enforce strict chronological separation across Training, Validation (Hyperparameter & Calibration Tuning), and Untouched Test partitions using purged and embargoed walk-forward cross-validation (Lopez de Prado, 2018) [14]. Every candidate model must be benchmarked against three mandatory parsimonious baselines: (a) a naive majority-class directional baseline (Always-UP), (b) a trailing 8-quarter issuer-specific historical average move baseline, and (c) the market-implied expected move extracted from pre-announcement at-the-money (ATM) straddle option prices [10, 11]. When evaluating multiple parameter trials across multiple cohorts, promotion gates must enforce the Deflated Sharpe Ratio (DSR) [2] and Benjamini-Hochberg FDR controls [27].

3. Computational and Agentic Scaling: Within the LangGraph / ReAct orchestrator [4, 7], deterministic statistical gates must govern stochastic LLM agents. The LLM critic's JSON hypotheses are treated strictly as candidate priors; no agent-proposed indicator or parameter change is retained unless it improves purged walk-forward validation metrics without violating complexity bounds.

4. Economic and Microstructure Scaling: Statistical forecast accuracy does not equate to economic profitability. Because option prices systematically embed elevated implied volatility ahead of scheduled earnings releases and experience rapid implied-volatility crush (IV crush) immediately upon announcement [10, 11], an equity move that is directionally correct can still lose money in long-option structures if the realized move falls short of the options-implied breakeven move. Therefore, the final promotion gate requires a minimum 30-day prospective paper-trading audit incorporating live bid-ask spreads, option chain implied-volatility surfaces, vega/theta decay, and capacity limits prior to any human-approved capital deployment.

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8. Conclusion

The primary contribution of the Autoresearch study is demonstrating that engineering an auditable, bias-free validation loop is more foundational than automating hypothesis generation. Our out-of-sample evaluation of 1,197 corporate earnings forecasts proves that the current baseline model does not possess a deployable directional (50.7% accuracy, 0.503 ROC AUC) or magnitude (8.60 percentage points MAE, -5.614 R-squared) edge. Crucially, however, the diagnostic harness transforms an uninformative aggregate failure into a precise engineering prescription: isolating upper-tail magnitude miscalibration (20.1% predicted vs. 6.0% realized in the top confidence quintile), identifying routing dilution from the pooled GLOBAL fallback model (48.4% accuracy across 50.7% of events), and flagging cohort-specific degradation in Biotechnology (41.1% accuracy) versus domain-specific promise in Software (54.5% accuracy). In quantitative finance, autonomous research systems must scale their scientific discipline and falsification rigor before scaling their autonomy or capital.

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